TeacherPoint — Bugs, Resolutions & SEO / AEO / GEO
Document version 1.0 · October 23, 2026 · Confidential — client delivery

TeacherPoint — Bugs, Resolutions & SEO / AEO / GEO Delivery Summary

Prepared for: Client share (development agreement alignment)

Product: TeacherPoint — public website + API

Repositories:

Component	GitHub	Primary branch
Frontend (Next.js 15)	`shivanshunigam01/website-hub-next`	`production-readiness` (deploy) · `fix/security-env-and-home-performance` (review)
Backend (Express / MongoDB)	`shivanshunigam01/teacheron-backend`	`production-readiness`

Agreement note: The signed agreement PDF is not stored in this repository. This document maps **delivered engineering work** to typical agreement themes: **production readiness**, **performance (Core Web Vitals)**, **SEO**, **AEO** (answer / AI discovery), and **GEO** (generative-engine optimization + **geographic** personalization). Where a milestone ID from your PDF differs, attach the PDF appendix and we can map row-by-row.

Live site: Changes below are **in Git**; production reflects them only after **host deploy** of the listed branches.

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1. Executive summary

Area	Status	Summary
Critical production bug (course page)	**Resolved in code**	Infinite `GET /courses/:id` loop fixed; deploy required on live
SEO (search engines)	**Strong baseline**	PSI SEO **100** (baseline); sitemap, robots, metadata, JSON-LD, hreflang
AEO / agentic discovery	**Implemented**	`llms.txt`, expanded machine-readable site map, FAQ/schema, factual copy
GEO (generative engines)	**Implemented**	`llms.txt` + structured data + truthful marketing copy (no fake stats)
GEO (geographic / regional)	**Improved**	Regional ads strip, deferred IP/geo, stable layout grid
Performance (CWV)	**Partial**	Major CLS/TBT/LCP mitigations shipped; **90+ performance not certified** without post-deploy PSI/GTmetrix
Security	**In progress**	Env files removed from Git tracking on feature branch; **rotation + history scrub = release blocker**

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2. Resolved bugs & issues (client-facing)

2.1 P0 — Production / user-facing

ID	Issue	Symptom	Root cause	Resolution	Verification
BUG-COURSE-001	Course detail API storm	Course URL shows endless spinner; **100+** identical API calls in browser Network tab	`useEffect` depended on entire LMS store; `loadCurriculum` updated store → infinite re-fetch loop	Removed LMS store from public course layout; **single** `useCourse()` query with in-flight dedupe; curriculum from API response	Commits on `production-readiness` (`1a790a1`, `0d7b3dc`); e2e `course-detail-api.spec.ts` (run against env with catalog data)
BUG-DEBUG-001	Console noise / failed requests	`POST http://127.0.0.1:7651/ingest/... ERR_CONNECTION_REFUSED`	Debug instrumentation left in production bundle	`debug-gt-log` no-op; ingest calls removed	No connection errors after deploy
BUG-SEO-001	Misleading hero claims	Unverified "120+ countries", "thousands of reviews"	Marketing copy on SSR hero	**Truthful, qualitative** hero copy on feature branch `HomeHeroServer`	Content review + legal sign-off recommended
BUG-PERF-001	GTmetrix regression (bad deploy)	Scores dropped after aggressive home refactor	Full-page hydrate + over-splitting sections	**Reverted**; restored deferred home chunk pattern (`production-readiness` `a3c3352`)	Re-test GTmetrix after deploy

2.2 P1 — Performance & layout (Core Web Vitals)

ID	Issue	Baseline (client-supplied PSI)	Mitigation	Expected impact
PERF-CLS-001	Desktop CLS ~**0.65**	Footer jumped when home sections hydrated	Below-fold height reservation, geo strip stability, section `min-h` shells, footer `min-h`	Lab desktop CLS **< 0.1** at 1280/1440 in Playwright (post-fix branch)
PERF-CLS-002	GTmetrix CLS ~**0.71**	Placeholder vs real grid mismatch; geo strip collapse	`GeoContentStrip` stable skeletons; grid geometry fix (`geo-content-grid.ts` on feature branch)	Re-measure GTmetrix after deploy
PERF-LCP-001	Mobile LCP ~**6.8s**	Heavy JS before hero paint	SSR hero **poster** (`next/image` priority) + static **H1**; tutor search deferred (`HeroSearchLazy`)	Confirm LCP element in PSI trace post-deploy (expected: poster or H1)
PERF-TBT-001	Desktop TBT ~**830ms**	Chat, translate, visitor, regional popup, geo IP on main thread	Tiered idle/scroll deferred widgets; route-scoped providers; home hero **no background MP4** (~500KB saved)	TBT improvement; not guaranteed <200ms without measurement

ID	Issue	Baseline (client-supplied PSI)	Mitigation	Expected impact
PERF-GEO-001	Geo strip layout shift	Loading grid used `limit` columns; loaded state used `shown.length`	Shared grid classes; single ad spans full row (`md:col-span-2`); no empty placeholder strip when zero ads	Reduced CLS on regional banners

2.3 P1 — Backend / API (production-readiness)

ID	Issue	Resolution
API-CACHE-001	Repeated anonymous course detail load	`Cache-Control` on public `GET /courses/:id` for anonymous clients (`014cb7c`)
API-SEC-001	Historical IDOR on users/notifications/payments	Hardening on `production-readiness` (per audit BR-005); **deploy + `test:security` on staging**

2.4 P1 — Auth & roles (frontend)

ID	Deliverable	Resolution
AUTH-001	Role-specific login entry	Public routes: `/login/student`, `/login/teacher`, `/login/parent`
AUTH-002	Staff / admin entry	`/staff-console` (hidden admin path) with role gating
AUTH-003	Navigation	Sign-up / nav links point to correct role logins

2.5 P1 — Architecture / bundle health

ID	Issue	Resolution
ARCH-001	Admin/LMS stores on every page	`RouteStoreProviders` — LMS/admin only on `/lms`, `/admin`, course detail gets `LearningProvider` only
ARCH-002	Google OAuth JS on all pages	`ScopedGoogleAuth` — `@react-oauth/google` only on auth routes (feature branch)
ARCH-003	Public home used admin mock catalog	`LearnAI`, `FeaturedTutors`, `IndustryExperts` use **`useCourses` / `useTutors`** (published API, shared cache) on feature branch

2.6 P0 — Security (feature branch — **client action required**)

ID	Issue	Resolution	**Client / ops action (blocker)**
SEC-ENV-001	`.env`, `.env.local`, `.env.production.local` were **tracked in public Git**	Removed from index; `.gitignore` updated; template remains `.env.example`	**Rotate** Geoapify, review Razorpay/Google keys; **purge secrets from Git history** — see `docs/security/ENV_ROTATION_AND_HISTORY.md`

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3. SEO (Search Engine Optimization) — delivered

Deliverable	Implementation	Client verification
Page metadata	`src/lib/page-metadata.ts`, per-route titles/descriptions/OG	View source / OG debugger on key URLs
Canonical URLs	`buildMetadata` / `SITE_URL`	No duplicate host variants
hreflang	`en` + `x-default` in metadata	International targeting baseline
robots.txt	`src/app/robots.ts` (dynamic)	Disallow staff/admin paths; allow public crawl
XML sitemap	`src/app/sitemap.ts` — static routes + **dynamic** tutor/course IDs + subject landing URLs	`https://www.teacherpoint.org/sitemap.xml` (~384+ URLs in audit)
Structured data (JSON-LD)	Global org/site in layout; **Course**, **Tutor**, **FAQ**, **Breadcrumb** components	Google Rich Results Test on course/tutor/FAQ pages
Server SEO helpers	`seo-server.ts` cached course/tutor fetches for metadata	Faster, consistent titles for detail pages
Popular subjects	Internal links + sitemap query URLs for tutor discovery	`/tutors?subject=…` entries intentional
PSI SEO score	Baseline **100** mobile/desktop (reported)	Re-run PSI after deploy

Script: `npm run seo:validate` (run against **production URL** with site up).

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4. AEO (Answer Engine Optimization) — delivered

AEO = content and structure so **answer engines** (Google AI Overviews, Bing Copilot, Perplexity, etc.) can cite accurate facts.

Deliverable	Details
FAQ content	Home + `/faq` with expandable Q&A; `FaqJsonLd` where applicable
Clear entity statements	About, contact, pricing, legal pages linked from footer and sitemap
Factual trust strip	`TrustStats` uses **qualitative** i18n strings (no unverified global counts)
Course/tutor detail	Server-rendered metadata + JSON-LD for named courses and tutors
Breadcrumbs	`BreadcrumbJsonLd` on course/tutor detail for hierarchy
Noindex on auth	Login/register/role-select excluded from marketing index where configured

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5. GEO — two meanings (both addressed)

5.1 GEO — Generative Engine Optimization (AI crawlers / LLMs)

Deliverable	Details
`/llms.txt`	`public/llms.txt` + `src/app/llms/route.ts`; rewrite in `next.config.ts`

Deliverable	Details
Plain-text summary	Who TeacherPoint is, what users can do, **no private repo paths**
15+ canonical links	Home, tutors, courses, workshops, jobs, marketplace, legal, sitemap
Content-Type	`text/plain; charset=utf-8` for agents
API pointer	Documented public API base for integrations (not HTML)
Agentic PSI	Baseline mobile **2/3**, desktop **1/3** agentic checks; layout stability fixes target remaining gap

5.2 GEO — Geographic / regional experience

Deliverable	Details
IP / location context	`LocationProvider`, deferred detect on non-critical routes
Regional ads	`GeoContentStrip`, `RegionalAdPopup` (deferred), `filterRegionalAds`
Currency / language	Currency + i18n + optional Google Translate (idle-loaded)
Geo API routes	Next.js `/api/geolocation/*` + backend `/api/v1/geo/ip` in production
Layout stability	Fixed grid geometry for 0/1/2 regional ads (feature branch)

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6. Performance & quality targets (honest status)

Target (agreement-style)	Baseline	Current engineering status
Mobile performance ≥ 90	~68 PSI	**Not certified** — optimizations merged; measure after deploy
Desktop performance ≥ 90	~46–47 PSI	**Not certified**
LCP ≤ 2.5s mobile	~6.8s	Mitigations in place; **PSI proof pending**
CLS ≤ 0.1	Mobile ~0.05; desktop ~0.65	Desktop **improved in lab**; mobile home handoff **~0.17** in Playwright on feature branch — **still tuning**
GTmetrix Grade A	~47% perf reported	**Not claimed** without new GTmetrix run
SEO 100	Yes (baseline)	Expected to hold; validate live

We do not claim Lighthouse/GTmetrix 90+ without post-deploy measurements.

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7. Automated tests & scripts

Test	Purpose	Typical result
`npm run build`	Production compile	**Pass** on current branches
`npx playwright test`	Smoke, home CLS, course API bound	**9 pass / 2 fail** on feature branch (mobile CLS threshold; course link needs live catalog)

Test	Purpose	Typical result
`npm run seo:validate`	robots, sitemap, llms, headers	Run against **live** URL
`npm run perf:lighthouse`	Local Lighthouse batch	Dev machine ≠ PSI
Backend `npm run test:security`	API auth regression	Run on **staging** after API deploy

8. Open items & remaining risks (transparent for client)

Priority	Item	Owner
P0	Rotate credentials + Git history cleanup (env files were public)	Client + DevOps
P0	Deploy `production-readiness` frontend + backend to production	Client hosting
P1	Mobile homepage CLS ≤ 0.1 (lab)	Engineering follow-up
P1	Admin console still uses mock data in places (`use-admin-store`)	Phase 0 audit backlog
P1	Paid enrollment without payment gate (product decision)	Product + backend
P1	Server-side admin RBAC (`requirePermission`) full enforcement	Backend
P2	NestJS/Postgres vs Express/Mongo contract clarification	Legal / PM
P2	npm audit (dev transitive advisories)	Engineering hygiene

Full internal register: `docs/production-audit/07_BUG_AND_RISK_REGISTER.md`.

9. Deployment checklist (for client ops)

1. Pull **`production-readiness`** on API server and Next.js host.
2. Set secrets **only** on server (never commit `.env`).
3. `npm ci && npm run build` (frontend).
4. Smoke: open `https://www.teacherpoint.org/courses/{valid-id}` → **one** course API call.
5. Smoke: `curl -I https://www.teacherpoint.org/llms.txt` → **200**, `text/plain`.
6. Run **PageSpeed Insights** (3× mobile, 3× desktop) — record medians.
7. Optional: GTmetrix / WebPageTest for grade tracking.
8. Merge **`fix/security-env-and-home-performance`** after security rotation + PR review.

10. Document control

Version	Date	Notes
1.0	2026-10-23	Initial client summary: bugs, SEO, AEO, GEO, performance, security

Attachments you may share with the client alongside this doc:

- `docs/performance/FINAL_DEPLOYMENT_CHECKLIST.md`
- `docs/security/ENV_ROTATION_AND_HISTORY.md`
- `docs/performance/FEATURE_BRANCH_REPORT.md` (technical appendix)
- `public/llms.txt` (live URL)

This summary describes engineering deliverables and honest verification status. It is not a legal amendment to the development agreement. For milestone sign-off, attach the signed PDF milestone table to section 2 rows.